



ATRIAL FIBRILLATION DETECTION SYSTEM USING DEEP LEARNING

A PROJECT REPORT

Submitted by

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ABSTRACT

Atrial Fibrillation (AF) is a prevalent cardiac arrhythmia that poses significant risks, including stroke and heart failure. Timely detection of AF is crucial for effective management and prevention of complications. This paper presents a novel system for AF detection using Convolutional Neural Networks (CNNs) applied to Electrocardiogram (ECG) signals. The proposed method involves transforming raw ECG data into 2D spectrograms to capture essential features indicative of AF. A CNN model is trained on these spectrograms to learn and identify patterns associated with AF episodes accurately. Experimental results demonstrate that the CNN-based approach achieves high sensitivity and specificity, outperforming traditional methods. The proposed system aims to enhance AF detection in real-time, providing a valuable tool for healthcare professionals and improving patient outcomes.

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LIST OF ABBREVIATIONS

YOLO	You Only Look Once
PDM	Plastic Detection Model
CNNs	Convolutional Neural Network
AI	Artificial Intelligence
RNNs	Recurrent Neural Network
GANs	Generative Adversarial Networks
ML	Machine Learning
DL	Deep Learning
NLP	Natural Language Processing
SRS	Software Requirement specification
FTIR	Fourier-Transform Infrared
UV	Ultra Violet Radiation
SVM	Support Vector Machine
CSV	Comma Separated Value
CLAHE	Contrast Limited Adaptive Histogram Equalization
PCA	Principal Component Analysis
SVD	Singular Value Decomposition

CHAPTER 1

INTRODUCTION

Atrial Fibrillation (AF) is one of the most common arrhythmias, affecting millions of individuals worldwide. Characterized by irregular and often rapid heart rhythms, AF can lead to severe complications, including stroke, heart failure, and increased mortality. Early detection is vital for effective treatment and management of AF, as timely intervention can significantly reduce the risk of adverse events. Traditional methods for AF detection rely on manual interpretation of ECG signals by healthcare professionals, which can be time-consuming and subject to human error. Recent advancements in machine learning, particularly Convolutional Neural Networks (CNNs), offer promising solutions for automating AF detection. CNNs, known for their ability to learn hierarchical features from images, can be effectively applied to ECG signals by converting them into 2D representations. This paper explores the application of CNNs for AF detection, detailing the methodology, results, and implications for real-time monitoring and diagnosis.

Atrial fibrillation (AFib) is a heart condition that causes the upper chambers of the heart to beat irregularly. It's the most common type of treated arrhythmia, which is an irregular heartbeat. Electrocardiogram (ECG) is a widely used diagnostic tool for monitoring heart activity and identifying cardiovascular diseases. AFib can cause an irregular heartbeat, fatigue, heart palpitations, trouble breathing, chest pain, dizziness, fainting, and low blood pressure. Detecting AF early is crucial as it increases the risk of stroke, heart failure, and other complications. It introduces a novel approach using Wavelet Transform and Convolutional Neural Networks (CNN) to detect AF from ECG signals.

Atrial fibrillation (AF) is the major concern with irregular heart rhythms, especially if a person approaches the age of 65. The heart is a pump used for efficient operation and it is regulated by an internal pacemaker that controls the heartbeat, the electrical impulse usually starts from the sinoatrial node and moves from atria to ventricles, which can cause regular rhythmic contractions of the chambers. AF risk remains unacknowledged in some patients, because of its unawareness, and other patients may be mindful of the erratic behaviour of heartbeat and may be uncomfortable with sensation. The irregular heartbeat indicates the symptoms of any accidental strokes that may result in further prolonged illness and, as a result, it leads to ultimate heart failure. The AF detection methods are mainly focused on the RR intervals, short term study of heart rate variability, and sequential examination to check the existence of P-wave. The current studies are mostly focused on the feature extraction techniques; therefore, the features have significant fictional effect on the final outcome of the models. The major improvements in electrocardiogram (ECG) data and the advancement of Dynamic Neural Network (DNNs) and, in particular, the implementation of algorithms such as long short term memory (LSTM) and convolutional neural network (CNN). These algorithms can be directly trained on the large-scaled dataset that RR Intervals of the ECG signal requirements and the performance of the ECG models is dramatically improved. It is worth mentioning that, in the past decade, deep learning (DL) classifiers have been successfully applied in different fields, such as speech recognition, image classification, and many other domains such as natural language processing.

However, DL algorithms have not been widely applied for AF detection. However, there are few studies applied aforementioned classifier for AF prediction, but the comparison results show that the performance was unsatisfactory when compared

to the image classification and image recognition. It is to be noted that there is not any study available to detect the AF using DL classifiers. In order to overcome the current issues, we build six models that are based on the feature-based approaches and DL approaches including support vector machine (SVM), multilayer perceptron (MLP), CNN and LSTM. The feature-based model is trained based on the manually extracted features, and DL methods are trained on raw data without any feature engineering. It is envisioned that this is the only exploiting automated feature engineering that focuses on DL for the automatic AF detection and classification. Additionally, the DL classifiers are compared with shallow learning classifiers.

Atrial Fibrillation (AF) is a common heart disease that is prevalence as increased age > 65 years old. Now, wearable medical devices are capable to be worn daily, so that, it can monitor our heart condition and alert if there is an abnormality. By connecting our medical devices to the smartphone that is commonly named mHealth, it is one example of digitalization in healthcare that can be done easily using wireless technology, such as Bluetooth, WI-Fi, etc.

So, to implement mHealth to the real-world application, we need to find new measurement techniques in our daily activities, to digitalize standard clinical medical device for data analysis, and to adapt the device to the ongoing regulations in clinical usage (Bhavnani et al., 2016). Another example is by using the iPhone with a single-lead ECG to perform a rapid AF screening by using an algorithm to detect the absence of the P-wave and manually interpret by the cardiologist (Lau et al., 2013).

Usually, to identify the AF, we can use the R-to-R interval analysis by calculating the absolute deviation and the difference between successive R-to-R intervals

(Ghodrati, Murray and Marinello, 2008). This method is commonly used by the smart-watch or wrist-band heart rate measurement device that is normally a fitness device

CHAPTER 2

LITERATURE REVIEW

DEEP LEARNING APPROACHES IN AF DETECTION- In recent years, deep learning techniques have gained traction in the detection of Atrial Fibrillation (AF). A landmark study by Rajpurkar et al. (2017) introduced a deep learning model that utilized convolutional neural networks (CNNs) to analyze raw ECG data for AF detection. This model demonstrated the ability to classify AF with high sensitivity and specificity, outperforming traditional cardiologist interpretations in certain instances. The authors highlighted the importance of large, annotated datasets for training the model, which allows CNNs to learn intricate patterns associated with AF. This research underscores the potential of deep learning to enhance diagnostic accuracy in clinical settings and emphasizes the need for integrating AI-driven solutions into routine cardiac care.

FEATURE EXTRACTION AND CLASSIFICATION TECHNIQUES- Asgari et al. (2015) explored various supervised learning algorithms, including Support Vector Machines (SVM), Random Forest (RF), and k-nearest neighbors (KNN), for AF detection. The study emphasized the importance of effective feature extraction techniques, which include both time-domain and frequency-domain analyses of ECG signals. Although traditional machine learning methods showed promise, they were often limited by their reliance on handcrafted features, which can overlook complex patterns in the data. This research pointed toward the need for advanced techniques like CNNs, which can automatically extract relevant features from raw ECG signals, enhancing classification performance and reducing the potential for human error.

WEARABLE TECHNOLOGY AND REAL-TIME MONITORING- Oster et al. (2018) investigated the application of machine learning algorithms in detecting AF using data from wearable devices, such as smartwatches and fitness trackers. This study demonstrated that data obtained from these devices could effectively identify AF episodes, providing a new avenue for remote monitoring and management of patients. The researchers utilized a combination of algorithms, including decision trees and ensemble methods, to classify heart rhythm abnormalities. The results suggest that wearable technology, when paired with robust machine learning models, can facilitate continuous monitoring and timely intervention, significantly improving patient outcomes in AF management.

CHALLENGES IN DATA QUALITY AND GENERALIZATION- Guglielmo et al. (2020) conducted a comprehensive analysis of ensemble learning techniques for AF detection and highlighted several challenges, including data quality, class imbalance, and generalization across diverse populations. The study showed that while ensemble methods could enhance classification performance, the effectiveness of these models heavily depended on the quality and diversity of the training data. Moreover, the authors pointed out the necessity of validation on external datasets to ensure the models' generalizability in clinical practice. This research emphasizes the critical need for high-quality, diverse datasets to train machine learning models for AF detection effectively.

TRANSFER LEARNING IN ECG ANALYSIS- Xu et al. (2023) explored the application of transfer learning techniques for AF detection, focusing on the adaptation of pre-trained models across different ECG datasets. Their research demonstrated that transfer learning could significantly enhance the performance of CNN models when applied to diverse populations and clinical settings. By leveraging knowledge gained from one dataset, the model could adapt to new,

unseen data more effectively, leading to improved classification accuracy. This study underscores the potential of transfer learning as a strategy to overcome data limitations in AF detection, facilitating the development of robust machine learning models that are both accurate and adaptable across various clinical environments.

The detection of Atrial Fibrillation using machine learning techniques has gained substantial attention in recent years. Several studies have explored various algorithms, highlighting the potential of deep learning methods, particularly CNNs, in improving diagnostic accuracy. For instance, Rajpurkar et al. (2017) introduced a deep learning model that demonstrated remarkable performance in AF detection from raw ECG data, showcasing the advantages of automated analysis over traditional approaches. Asgari et al. (2015) compared supervised learning algorithms such as Support Vector Machines and Random Forests, but they noted limitations in feature extraction and classification accuracy, underscoring the need for advanced techniques.

The use of CNNs in ECG analysis has proven to be a transformative approach. Yildirim et al. (2019) applied CNNs to detect AF by transforming ECG signals into images, significantly improving classification performance compared to traditional methods. This study emphasized the importance of preprocessing techniques, such as noise filtering and segmentation, which enhance the quality of the input data. Additionally, Attia et al. (2019) developed a CNN model capable of processing long-term ECG recordings, highlighting the model's robustness in real-world scenarios.

Recent literature also underscores the challenges in AF detection, including data imbalance and generalizability across different populations. Guglielmo et al.

(2020) conducted a comparative analysis of ensemble learning techniques, emphasizing the need for diverse datasets to improve model accuracy. Furthermore, Xu et al. (2023) explored transfer learning to enhance the CNN's performance on varied ECG datasets, demonstrating that pre-trained models could effectively adapt to different clinical environments.

Overall, the literature indicates that while traditional methods for AF detection have made significant strides, the integration of CNNs offers a promising pathway toward more accurate, scalable, and automated solutions. This paper builds on these foundations by presenting a comprehensive approach that leverages CNNs for real-time AF detection, aiming to advance the capabilities of existing diagnostic tools in clinical settings.

CHAPTER 3

EXISTING SYSTEM

Access to quality healthcare for heart diseases remains a significant challenge in developing countries like Indonesia, primarily due to its diverse geographical landscape. Heart disease is prevalent not only in urban centers but also in rural areas, driven by unhealthy lifestyles and dietary habits. Given that heart diseases often exhibit gradual changes in symptoms, continuous monitoring of cardiac health is essential. This monitoring can be achieved through the use of wearable medical devices capable of tracking heart conditions and alerting users to abnormalities.

Current embedded systems include wearable devices that can be comfortably worn on the chest, which are designed for real-time data acquisition and processing of electrocardiogram (ECG) signals. These devices typically comprise a 1-lead ECG sensor, an ARM processor, a Bluetooth module for data transmission, an SD card for storage, and rechargeable batteries for portability.

By employing digital filtering techniques and the Tompkins algorithm, these systems can effectively identify P-wave presence and calculate heart rate variability metrics, such as heartbeat count, average heartbeat, standard deviation, and root mean square values. To facilitate the pre-diagnosis of atrial fibrillation, these systems leverage artificial neural networks (ANN) featuring 4 input neurons, 6 hidden neurons, and 1 output neuron. The network is structured using multi-layer perceptrons and trained through backpropagation.



Fig.3.1 Atrial Fibrillation Techniques

In practical applications, this approach has been validated using 41 recorded training samples from the PhysioNet MIT/BIH Atrial Fibrillation Database and Normal Sinus Rhythm Database, alongside test samples from 6 healthy individuals. The ANN achieved a remarkably low error rate of 0.1%, necessitating 31,548 epochs for optimal training and classification of heart conditions.

Given these advancements, such prototypes represent a viable solution for medical-grade wearable devices that can aid cardiologists in providing early warnings about users' heart health. By implementing these systems in rural areas, there is potential to significantly reduce the risk of sudden cardiac events and enhance overall health outcomes for populations at risk of heart disease.

The embedded device has a single-lead ECG with analog filters to pass only the ECG signal and to be acquired by a 10-bit analog-to-digital converter on the ARM Cortex-M4 microprocessor at 200 samples/sec, then the ECG signals will be extracted every 3 seconds to retrieve the heart rate and P-wave presences. Next, the heart rate variability values are computed to result the ANN, SDANN, and RMSSD that is related to the irregularity of the heart rate as one of the AF parameters.

Also, the P-wave presences is calculated every 3 seconds to find the percentage of the P-wave appearance that is related with the fibrillation at the heart's atrial. Both, of these parameters are used to determine AF. Previously, artificial intelligence has been used for arrhythmias with artificial neural network (Modarressi et al., 2016) and convolution neural network (Rajpurkar et al., 2017), however, deep learning (Yuan et al., 2016; Shashikumar et al., 2017; Gotlibovych et al., 2018) has been used to for AF classification from the heart rate.

However, in this research we have implemented an artificial neural network on an embedded device to classify AF from irregular heart rate and P-wave presences percentage. So, this prototype can be used as a medical-grade wearable device that can assist or even give help to the cardiologist by providing the AF knowledge to the embedded device and pre-diagnose the user's heart condition.

CHAPTER 4

PROPOSED SYSTEM

The proposed system for Atrial Fibrillation (AF) detection combines Convolutional Neural Networks (CNNs) applied to ECG signals to automate the diagnosis process and enhance accuracy. The system begins with data acquisition through wearable ECG monitors that continuously collect patient data. This real-time ECG data undergoes a preprocessing phase, where noise and artifacts are removed using advanced filtering techniques, and the signals are segmented into manageable intervals. Each ECG segment is then transformed into a 2D spectrogram, which captures the frequency components of the signal, making it suitable for analysis by a CNN.

The CNN model is trained on a diverse dataset of spectrograms, enabling it to learn complex patterns associated with AF episodes. The training process includes validation techniques to assess model performance and mitigate overfitting. Once trained, the model can classify incoming ECG segments in real-time, identifying AF episodes with high sensitivity and specificity. The system is equipped with a user-friendly interface that presents ECG data and classification results to both patients and healthcare providers, facilitating timely intervention. Additionally, the system can be integrated with cloud storage for secure data management and remote access, ensuring continuous monitoring and support for patients. Overall, this proposed system aims to provide a reliable, efficient, and scalable solution for AF detection, enhancing the standard of care in cardiac health.

4.1 BLOCK DIAGRAM:

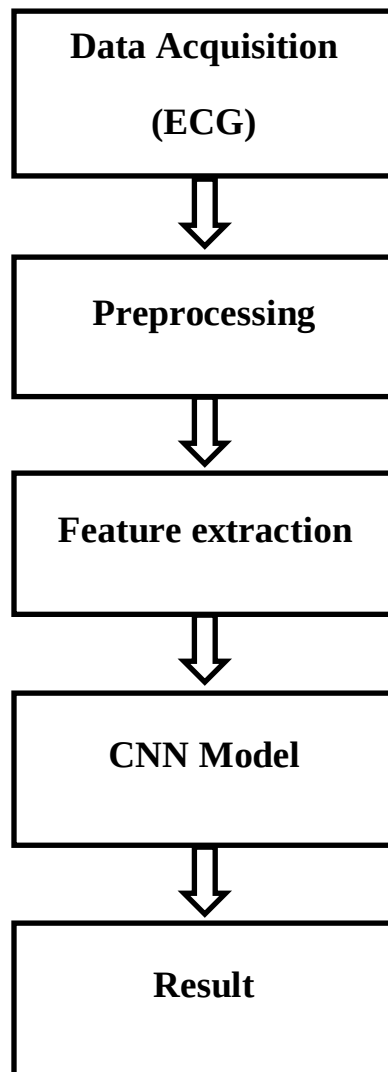


Fig.4.1 – Block Diagram

CHAPTER 5

SYSTEM REQUIREMENTS

5.1 HARDWARE REQUIREMENTS

- Processor: intel Core i7 or AMD Ryzen 7.
- GPU: NVIDIA GPU with CUDA support (e.g., GTX 1080 Ti, RTX 2080, or higher).
- Memory (RAM): Minimum 16 GB
- Storage: SSD with at least 1 TB

5.2 SOFTWARE REQUIREMENTS

- Operating System: Windows 10/11, macOS, or Linux.
- Python Environment: Python 3.2 or higher.
- Libraries and Frameworks:
 - TensorFlow or Keras (for implementing and fine-tuning the VGG19 model).
 - OpenCV (for image preprocessing tasks such as resizing and normalization).
 - NumPy (for numerical operations and data manipulation).
 - Scikit-learn (for evaluation metrics and dataset splitting).

- CUDA Toolkit and cuDNN (if using an NVIDIA GPU) for GPU acceleration.
- Jupyter Notebook (for developing, testing, and demonstrating the system).

5.3 SYSTEM REQUIREMENT SPECIFICATION

This chapter describes about the requirements. It specifies the hardware and software requirements that are required in order to run the application properly. The Software Requirement Specification (SRS) is explained in detail, which includes overview of dissertation as well as the functional and non-functional requirement of this dissertation.

A SRS document describes all data, functional and behavioural requirements of the software under production or development. SRS is a fundamental document, which forms the foundation of the software development process. It is the complete description of the behaviour of a system to be developed. Requirement Analysis discusses the conditions to be met for a new or altered product. Requirement Analysis is critical to the success to a development project. Requirement must be documented, measurable, testable, related to in identified business needs or opportunities, and defined to a level of detail sufficient for system design. The SRS functions as a blueprint for completing a project. The goal of preparing the SRS document is to:

- Facilitate communication between the customer, analyst, system developers, maintainers.
- To form a foundation for the design phase.
- Support system testing facilities.
- Controlling the evolution of the system.



Fig. 5.3 Python Modules

- **Python**

Python is a general-purpose interpreted, interactive, object-oriented, and high-level programming language. Python is designed to be highly readable. It uses English keywords frequently where as other languages use punctuation, and it has fewer syntactical constructions than other languages.

- **Scikit-learn:**

It is one of the most popular libraries for machine learning in Python. It provides simple yet efficient tools for data analysis and modelling. Scikit-learn supports a wide range of supervised learning algorithms like linear regression, decision trees, support vector machines (SVMs), and random forests, as well as unsupervised learning methods such as clustering (K-means, DBSCAN) and dimensionality reduction (PCA). It also includes utilities for model evaluation, offering metrics like accuracy, precision, recall, and cross-validation. Additionally, Scikit-learn provides various preprocessing techniques, such as data normalization, scaling, and encoding, making it ideal for building, training, and evaluating machine learning models.

- **Tensorflow:**

It is a powerful open-source deep learning framework developed by Google, widely used for developing and deploying machine learning and neural network models. TensorFlow allows for efficient multi-dimensional array (tensor) computations and provides the necessary tools for creating and training deep learning models such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Generative

Adversarial Networks (GANs). Its high-level API, Keras, simplifies the process of building models. TensorFlow supports distributed computing, enabling model training across multiple GPUs or machines, making it perfect for large-scale deep learning applications, such as image classification, natural language processing (NLP), and object detection.

- **NumPy:**

It is the fundamental package for numerical and scientific computing in Python. It provides support for large, multi-dimensional arrays and matrices, along with an extensive collection of mathematical functions to operate on these arrays. NumPy is efficient at handling numerical computations, and its operations are often used as building blocks for more complex data manipulation tasks. It supports basic arithmetic, statistical functions, and linear algebra operations, along with broadcasting for performing operations on arrays of different shapes. NumPy's array handling is the foundation for many other libraries like TensorFlow and Scikit-learn, making it essential for machine learning and scientific computing projects.

- **Pandas:**

It is a highly versatile library for data manipulation and analysis. It provides powerful data structures like DataFrames, which allow for easy handling and manipulation of tabular data (structured data). Pandas is great at data cleaning, providing functionality to handle missing values, filter and transform data, and perform operations like grouping and aggregating data. With Pandas, data from different sources like CSV files, Excel sheets, SQL databases, and JSON formats can be easily loaded and processed. Its flexible and fast handling of data makes it the go-to library for data scientists when working on data analysis and preparation tasks.

- **OpenCV:**

OpenCV (Open Source Computer Vision) is a popular library for real-time image processing and computer vision applications. It provides a wide range of functions for tasks like image manipulation (resizing, filtering, and transformations), edge detection, and object recognition. OpenCV is widely used in both academic and industrial applications, particularly for real-time video analysis and image recognition tasks. It

supports video processing, enabling the capture and manipulation of video streams for applications like motion detection and object tracking. OpenCV can also be integrated with deep learning models, making it a robust tool for advanced computer vision tasks such as object detection and facial recognition.

- **GUI**

Python supports GUI applications that can be created and ported to many system calls, libraries and windows systems, such as Windows MFC, Macintosh, and the X Window system of UNIX.

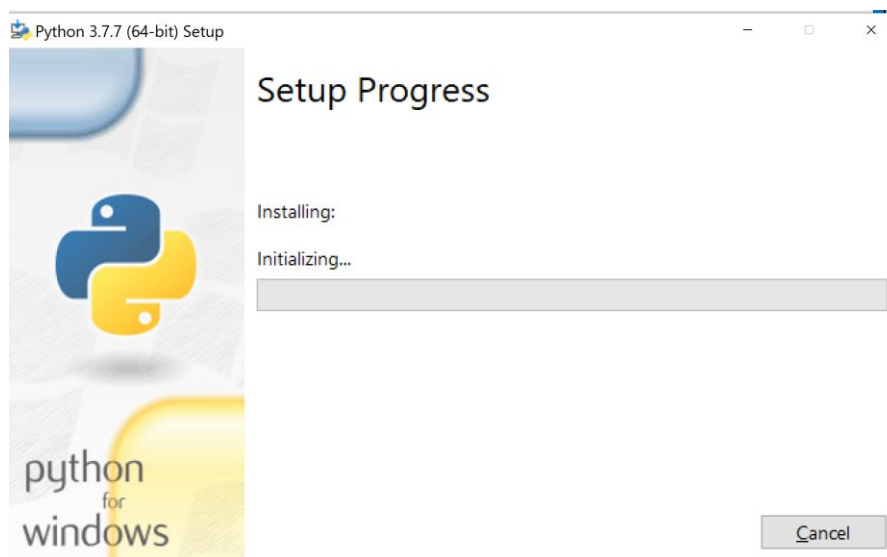


Fig. 5.3.1 Python for Windows Setup

5.3.1 NON-FUNCTIONAL REQUIREMENTS:

Non-functional requirements are the requirements which are not directly concerned with the specific function delivered by the system. They specify the criteria that can be used to judge the operation of a system rather than specific behaviours. They may relate to emergent system properties. Non-functional requirements are requirements that are not specifically concerned with the functionality of a system. They normally place restrictions on the product being developed and the development process. Non-functional requirements may be regarded as parameters of functionality in that they determine how quickly, how accurately, how reliably,

how securely, etc., functions must operate. Some of the ALPR non-functional requirements are as follows:

- The system may issue a receipt to remove any papers printed and make it a green initiative
- The system must be working at 100% peak efficiency
- When checking the database for errors, a 100% scan of the data is required, rather than selecting a sample set
- A process must be devised to support normal precinct business hours
- The system should provide documentation to inform users of system functionality and any change to the system
- The system should provide friendly graphical Interface to ensure ease of use when end users utilize system functionality.

5.3.2 CONVOLUTED NEURAL NETWORK (CNN):

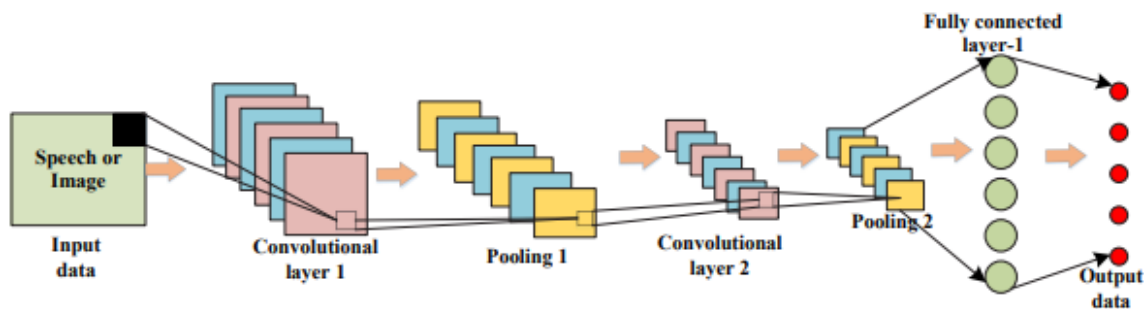


Fig. 5.3.2 Convolutional Neural Network (CNN)

A Convolutional Neural Network (CNN) is a type of deep learning algorithm that is particularly well suited for image processing tasks. It is made up of multiple layers, including convolutional layers, pooling layers, and fully connected layers. The convolutional layers are the key component of a CNN, where filters are applied to the input image to extract features such as edges, textures, and shapes. The output of the convolutional layers is then passed through

pooling layers, which are used to down-sample the feature maps, reducing the spatial dimensions while retaining the most important information. The output of the pooling layers is then passed through one or more fully connected layers, which are used to make a prediction or classify the image. CNNs are trained using a large dataset of labelled images, where the network learns to recognize patterns and features that are associated with specific objects or classes. Once trained, a CNN can be used to classify new images, or extract features for use in other applications such as object detection or image segmentation. CNNs have achieved state-of-the-art performance on a wide range of image recognition tasks, including object classification, object detection, and image segmentation. They are widely used in computer vision, image processing, and other related fields, and have been applied to a wide range of applications, including self-driving cars, medical imaging, and security systems.

Convolutional Neural Network Design:

- The construction of a convolutional neural network is a multi-layered feed-forward neural network, made by assembling many unseen layers on top of each other in a particular order.
- It is the sequential design that give permission to CNN to learn hierarchical attributes.
- In CNN, some of them followed by grouping layers and hidden layers are typically convolutional layers followed by activation layers.
- The pre-processing needed in a ConvNet is kindred to that of the related pattern of neurons in the human brain and was motivated by the organization of the Visual Cortex.

Different Types of CNN Models:

- LeNet
- AlexNet
- ResNet
- GoogleNet
- MobileNet
- VGG

OpenCV:

OpenCV was started at Intel in 1999 by Gary Bradsky, and the first release came out in 2000. Vadim Pisarevsky joined Gary Bradsky to manage Intel's Russian software OpenCV team.

In 2005, OpenCV was used on Stanley, the vehicle that won the 2005 DARPA Grand Challenge. Later, its active development continued under the support of Willow Garage with Gary Bradsky and Vadim Pisarevsky leading the project. OpenCV now supports a multitude of algorithms related to Computer Vision and Machine Learning and is expanding day by day. OpenCV supports a wide variety of programming languages such as C++, Python, Java, etc., and is available on different platforms including Windows, Linux, OS X, Android, and iOS. Interfaces for high-speed GPU operations based on CUDA and OpenCV are also under active development.

OpenCV-Python is the Python API for OpenCV, combining the best qualities of the OpenCV C++ API and the Python language. OpenCV-Python is a library of Python bindings designed to solve computer vision problems. Python is a general purpose programming language started by Guido van Rossum that became very popular very quickly, mainly because of its simplicity and code readability. It enables the programmer to express ideas in fewer lines of code without reducing readability.

Compared to languages like C/C++, Python is slower. That said, Python can be easily extended with C/C++, which allows us to write computationally intensive code in C/C++ and create Python wrappers that can be used as Python modules. This gives us two advantages: first, the code is as fast as the original C/C++ code (since it is the actual C++ code working in background) and second, it is easier to code in Python than C/C++. OpenCV-Python is a Python wrapper for the original OpenCV C++ implementation. OpenCV-Python makes use of Numpy, which is a highly optimized library for numerical operations with MATLAB-style syntax. All the OpenCV array structures are converted to and from Numpy arrays. This also makes it easier to integrate with other libraries that use Numpy such as SciPy and Matplotlib. Applications of OpenCV: There are lots of applications which are solved using OpenCV; some of them are listed below

- Face recognition
- Automated inspection and surveillance
- Number of people – count (foot traffic in a mall, etc)
- Vehicle counting on highways along with their speeds

- Interactive art installations
- Anomaly (defect) detection in the manufacturing process (the odd defective products)
- Street view image stitching
- Video/image search and retrieval
- Robot and driver-less car navigation and control
- object recognition
- Medical image analysis
- Movies – 3D structure from motion
- TV Channels advertisement recognition

OpenCV Functionality

- Image/video I/O, processing, display (core, imgproc, highgui)
- Object/feature detection (objdetect, features2d, nonfree)
- Geometry-based monocular or stereo computer vision (calib3d, stitching, videostab)
- Computational photography (photo, video, superres)
- Machine learning & clustering (ml, flann)
- CUDA acceleration (gpu)

Image-Processing:

Image processing is a method to perform some operations on an image, in order to get an enhanced image and or to extract some useful information from it. If we talk about the basic definition of image processing then “Image processing is the analysis and manipulation of a digitized image, especially in order to improve its quality”.

Digital-Image:

An image may be defined as a two-dimensional function $f(x, y)$, where x and y are spatial (plane) coordinates, and the amplitude of fat any pair of coordinates (x, y) is called the intensity or grey level of the image at that point. In another word An image is nothing more than a two-dimensional matrix (3-D in case of colored images) which is defined by the mathematical function $f(x, y)$ at any point is giving the pixel value at that point of an image, the pixel value describes how bright that pixel is, and what color it should be.

Image processing is basically signal processing in which input is an image and output is image or characteristics according to requirement associated with that image. Image processing basically includes the following three steps:

1. Importing the image
2. Analysing and manipulating the image
3. Output in which result can be altered image or report that is based on image analysis

Haar Cascade Algorithm:

It is an Object Detection Algorithm used to identify faces in an image or a real time video. The algorithm uses edge or line detection features proposed by Viola and Jones in their research paper “Rapid Object Detection using a Boosted Cascade of Simple Features” published in 2001. The algorithm is given a lot of positive images consisting of faces, and a lot of negative 22 images not consisting of any face to train on them. The model created from this training is available at the OpenCV GitHub repository.

Feature:

The first contribution to the research was the introduction of the haar features. These features on the image make it easy to find out the edges or the lines in the image, or to pick areas where there is a sudden change in the intensities of the pixels.

The objective here is to find out the sum of all the image pixels lying in the darker area of the haar feature and the sum of all the image pixels lying in the lighter area of the haar feature. And

then find out their difference. Now if the image has an edge separating dark pixels on the right and light pixels on the left, then the haar value will be closer to 1. That means, we say that there is an edge detected if the haar value is closer to 1. In the example above, there is no edge as the haar value is far from 1.

This is just one representation of a particular haar feature separating a vertical edge. Now there are other haar features as well, which will detect edges in other directions and any other image structures. To detect an edge anywhere in the image, the haar feature needs to traverse the whole image.

The haar feature continuously traverses from the top left of the image to the bottom right to search for the particular feature. This is just a representation of the whole concept of the haar feature traversal. In its actual work, the haar feature would traverse pixel by pixel in the image. Also all possible sizes of the haar features will be applied.

Depending on the feature each one is looking for, these are broadly classified into three categories. The first set of two rectangle features is responsible for finding out the edges in a horizontal or in a vertical direction (as shown above). The second set of three rectangle features are responsible for finding out if there is a lighter region surrounded by darker regions on either side or vice-versa. The third set of four rectangle features is responsible for finding out change of pixel intensities across diagonals.

Now, the haar features traversal on an image would involve a lot of mathematical calculations. As we can see for a single rectangle on either side, it involves 18 pixel value additions (for a rectangle enclosing 18 pixels). Imagine doing this for the whole image with all sizes of the haar features. This would be a hectic operation even for a high performance machine.

To tackle this, they introduced another concept known as The Integral Image to perform the same operation. An Integral Image is calculated from the Original Image in such a way that each pixel in this is the sum of all the pixels lying in its left and above in the Original Image. The calculation of a pixel in the Integral Image can be seen in the above GIF. The last pixel at the bottom right corner of the Integral Image will be the sum of all the pixels in the Original Image.

Training Set Selection: Each weak classifier should be trained on a random subset of the total training set. The subsets can overlap—it's not the same as, for example, dividing the training set into ten portions. AdaBoost assigns a “weight” to each training example, which determines the probability that each example should appear in the training set. Examples with higher weights are more likely to be included in the training set, and vice versa. After training a classifier, AdaBoost increases the weight on the misclassified examples so that these examples will make up a larger part of the next classifiers training set, and hopefully the next classifier trained will perform better on them. The equation for this weight update step is detailed later on.

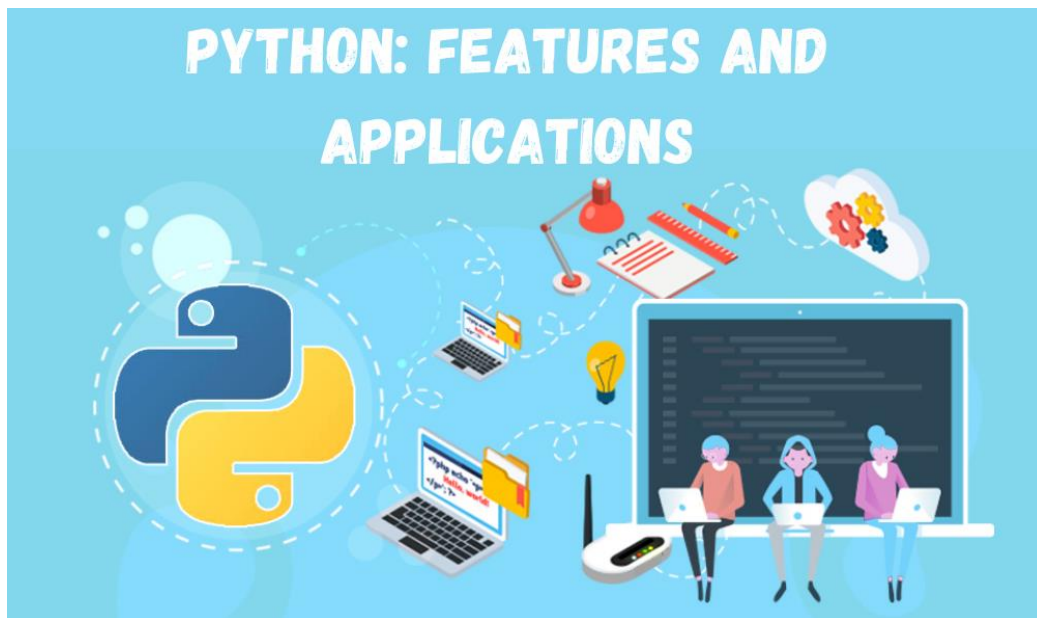


Fig.5.3.1.1 Python: Features and Applications

Classifier Output Weights:

After each classifier is trained, the classifier’s weight is calculated based on its accuracy. More accurate classifiers are given more weight. A classifier with 50% accuracy is given a weight of zero, and a classifier with less than 50% accuracy (kind of a funny concept) is given negative weight.

5.3.3 WHY PYTHON:

Python is a widely used high-level, general-purpose, interpreted, dynamic programming language. Its design philosophy emphasizes code readability, and its syntax allows programmers to express concepts in fewer lines of code than would be possible in languages such as C++ or Java. The language provides constructs intended to enable clear programs on both a small and large scale. Python supports multiple programming paradigms, including object-oriented, imperative and functional programming or procedural styles. It features a dynamic type system and automatic memory management and has a large and comprehensive standard library. Python interpreters are available for installation on many operating systems, allowing Python code execution on a wide variety of systems.

Python provides many useful features to the programmer. These features make it the most popular and widely used language. We have listed below few-essential features of Python.

- **Easy to use and Learn:** Python has a simple and easy-to-understand syntax, unlike traditional languages like C, C++, Java, etc., making it easy for beginners to learn.
- **Expressive Language:** It allows programmers to express complex concepts in just a few lines of code or reduces Developer's Time.
- **Interpreted Language:** Python does not require compilation, allowing rapid development and testing. It uses Interpreter instead of Compiler.

Object-Oriented Language: It supports object-oriented programming, making writing reusable and modular code easy.

- **Open Source Language:** Python is open source and free to use, distribute and modify.
- **Extensible:** Python can be extended with modules written in C, C++, or other languages.
- **Learn Standard Library:** Python's standard library contains many modules and functions that can be used for various tasks, such as string manipulation, web programming, and more.
- **GUI Programming Support:** Python provides several GUI frameworks, such as Tkinter and PyQt, allowing developers to create desktop applications easily.

- **Integrated:** Python can easily integrate with other languages and technologies, such as C/C++, Java, and .NET.
- **Embeddable:** Python code can be embedded into other applications as a scripting language.
- **Dynamic Memory Allocation:** Python automatically manages memory allocation, making it easier for developers to write complex programs without worrying about memory management.
- **Wide Range of Libraries and Frameworks:** Python has a vast collection of libraries and frameworks, such as NumPy, Pandas, Django, and Flask, that can be used to solve a wide range of problems.
- **Versatility:** Python is a universal language in various domains such as web development, machine learning, data analysis, scientific computing, and more.
- **Large Community:** Python has a vast and active community of developers contributing to its development and offering support. This makes it easy for beginners to get help and learn from experienced developers.
- **Career Opportunities:** Python is a highly popular language in the job market. Learning Python can open up several career opportunities in data science, artificial intelligence, web development, and more.
- **High Demand:** With the growing demand for automation and digital transformation, the need for Python developers is rising. Many industries seek skilled Python developers to help build their digital infrastructure.
- **Increased Productivity:** Python has a simple syntax and powerful libraries that can help developers write code faster and more efficiently. This can increase productivity and save time for developers and organizations.
- **Big Data and Machine Learning:** Python has become the go-to language for big data and machine learning. Python has become popular among data scientists and machine learning engineers with libraries like NumPy, Pandas, Scikit-learn, TensorFlow, and more.

Python Popular Frameworks and Libraries:

Python has wide range of libraries and frameworks widely used in various fields such as machine learning, artificial intelligence, web applications, etc. We define some popular frameworks and libraries of Python as follows.

- Web development (Server-side) - Django Flask, Pyramid, CherryPy
- GUIs based applications - Tk, PyGTK, PyQt, PyJs, etc.
- Machine Learning - TensorFlow, PyTorch, Scikit-learn, Matplotlib, Scipy, etc.
- Mathematics - Numpy, Pandas, etc.
- BeautifulSoup: a library for web scraping and parsing HTML and XML
- Requests: a library for making HTTP requests
- SQLAlchemy: a library for working with SQL databases
- Kivy: a framework for building multi-touch applications
- Pygame: a library for game development
- Pytest: a testing framework for Python
- Django REST framework: a toolkit for building RESTful APIs
- FastAPI: a modern, fast web framework for building APIs
- Streamlit: a library for building interactive web apps for machine learning and data science
- NLTK: a library for natural language processing.

Scripting Language:

A scripting or script language is a programming language that supports scripts, programs written for a special run-time environment that automate the execution of tasks that could alternatively be executed one-by-one by a human operator. Scripting languages are often interpreted (rather than compiled). Primitives are usually the elementary tasks or API calls, and the language allows them to be combined into more complex programs. Environments that can be automated through scripting include software applications, web pages within a web browser, the shells of operating

systems (OS), embedded systems, as well as numerous games. A scripting language can be viewed as a domain-specific language for a particular environment; in the case of scripting an application, this is also known as an extension language. Scripting languages are also sometimes referred to as very high-level programming languages, as they operate at a high level of abstraction, or as control languages.

Object Oriented Programming Language:

Object-oriented programming (OOP) is a programming paradigm based on the concept of "objects", which may contain data, in the form of fields, often known as attributes; and code, in the form of procedures, often known as methods. A distinguishing feature of objects is that an object's procedures can access and often modify the data fields of the object with which they are associated (objects have a notion of "this" or "self"). In OO programming, computer programs are designed by making them out of objects that interact with one another. There is significant diversity in object oriented programming, but most popular languages are class-based, meaning that objects are instances of classes, which typically also determines their type.

Data Pre-Processing:

For the CBIS-DDSM and INbreast datasets, the coordinates of the ROIs bounding box required for Yolo training were calculated considering the coordinates of the smallest rectangle containing the segmented lesion. Instead, the ROI coordinates for the proprietary dataset were computed from the square region that inscribes the circle containing the ROI. The CBIS-DDSM dataset has an acceptable size for deep learning architecture training.

However, it is composed of scanned film mammograms, much noisier and less detailed than FFDM. For this reason, only for the CBIS-DDSM dataset, the contrast limited adaptive histogram equalization (CLAHE) was applied for image enhancement, with the following setting: 1 as contrast limit, 2×2 as grid size, followed by a 3×3 Gaussian filter. For all data-sets, the gray levels were scaled in the range 0–255, and the images were resized to 640×640 using the Lanczos filter. The CBIS-DDSM dataset was spitted randomly considering 70% training, 15% validation, and 15% test set. Conversely, the INbreast and the proprietary datasets were split into training (80%) and test set (20%), respectively. Considering the small size of the two datasets and the unbalanced issue, the next “Data Augmentation” discusses data augmentation

for class balancing and generation of the validation set “Techniques for Class Balancing Before the Training Phase”, as well as the procedure to improve the training “Techniques Used During the Training Phase”.

Data Augmentation Techniques for Class Balancing Before the Training Phase. Due to the excessive imbalance classes for the INbreast and proprietary dataset, the minority class images (benign) of the training set were augmented. Although the main purpose of the work is to evaluate the detection performance on the proprietary dataset (regardless of lesion class), the following data augmentation procedure was applied to the proprietary dataset before the training phase. Figure 2 summarizes the transformation considered. In particular, 180° rotation and 180° rotation + flip upper-down (UD) were applied for benign images. The other transformations were applied during the training of Yolo, as discussed in the next subsection “Techniques Used during the Training Phase”.

In addition, according to, the remaining test dataset was augmented to obtain the validation set. In fact, flip UD, 180° rotation + flip UD, flip left-right (LR) and 180° rotation were applied on benign images, and Flip LR for malignant images. Considering the smaller difference between the classes, on INbreast, also, 180° rotations for malignant masses was considered [9]. This procedure resulted in the generation of a balanced validation set. In addition, the discussed procedure for INbreast and the proprietary datasets was repeated considering 5 different splitting of training and test sets (5-fold cross-validation).

Yolo Architectures Training:

Like other single-stage object detectors, Yolo consists of three parts: backbone, neck, and head. The backbone part is a CNN that extracts and aggregates image features. The neck part allows for features extraction optimized for small, medium, and large object detection. In the end, the three feature maps for small, medium, and large object detection are given as input to the head part, thus composed of convolutional layers for the final prediction. Yolo requires that the image is divided into a grid, and then makes a prediction for each grid cell. The prediction consists of a 6-tuple $y=(pc, bx, by, bh, bw, c)$, where (bx, by, bh, bw) identify coordinates (x, y) and sizes (height, width) of the predicted bounding-box, pc represent the probability that there is an object in the cell, and c represent the predicted class. The mechanism of anchors is also used, to allow multiple object detection in the same grid cell. For this reason, the prediction is the 6-tuple

discussed for each specified anchor. Each version of Yolo has its own peculiarities, which mainly concern the structure of the feature extractor, that is, the backbone.

YoloV3 Model:

YoloV3 is much deeper than the previous two versions and is more accurate but requires more time and data for training. In YoloV3, the Darknet53 was used as backbone. Darknet53 is a hybrid approach between Darknet19 (used in YoloV2) and residual network elements (e.g., BottleNeck), proposed to improve the Darknet19 and the efficiency of ResNet-101/152. The short-cut connections allow getting more fine-grained information, leading to better performance for small objects. The feature pyramids network (FPN) is used as neck, allowing learning objects of different sizes: it specializes in detecting large and small objects. In addition, the non-maximum suppression to select one bounding box out of many overlap-ping bounding boxes is used. YoloV5 Model YoloV5 uses CSPDarknet53 as its backbone: it exploits the architecture Darknet53 proposed in and employs a CSPNet strategy to partition the feature map of the base layer into two parts and then merges them through a cross-stage hierarchy. In the neck part, PAnet is used to generate the feature pyramids network (FPN) and allow the extraction of multi-scale feature maps. This structure allows the extraction of features optimized for small, medium, and large object detection. YoloV5 was released in nano, small, medium, large, and extra-large versions. The versions differ in the number of convolutional kernels used and thus the number of parameters. In this paper, a comparison between nano, small, medium, and large versions was performed.

YoloV5-Transformer:

In contrast to convolutional networks, Transformer are able to model the relationships among various small patches in the image. The Transformer block assumes the image is split into a sequence of patches, where each patch is flattened to a vector. These flattened image patches are used to create lower-dimensional linear embeddings and fed into a Transformer encoder, composed by a multi-head attention to find local and global dependencies in the image. It has been shown that the introduction of a Transformer block to convolutional networks can improve efficiency and overall accuracy. In YoloV5, the Transformer block was embedded in the penultimate layer of the backbone, that is, among the three convolutional layers preceding the spatial pyramid pooling layer.

Python Features:

Python provides many useful features which make it popular and valuable from the other programming languages. It supports object-oriented programming, procedural programming approaches and provides dynamic memory allocation. We have listed below a few essential features.

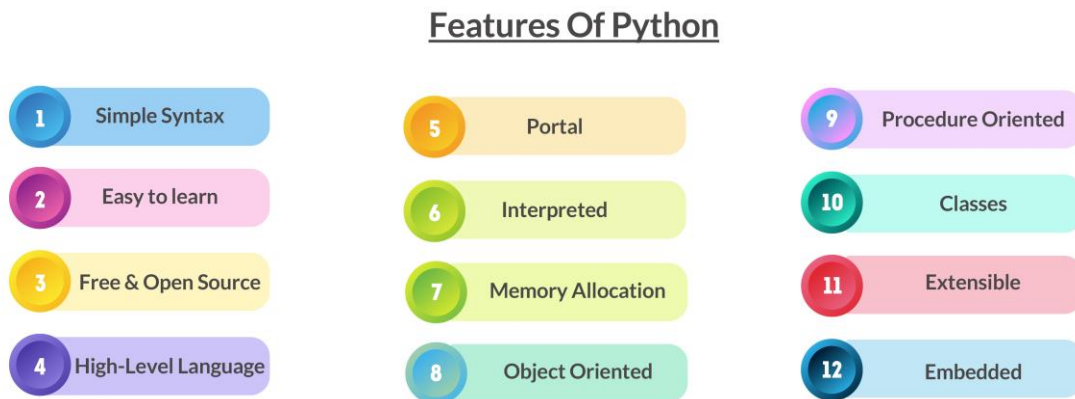


Fig.5.3.1.2 Features of Python

1) Easy to Learn and Use

Python is easy to learn as compared to other programming languages. Its syntax is straightforward and much the same as the English language. There is no use of the semicolon or curly-bracket, the indentation defines the code block. It is the recommended programming language for beginners.

2) Expressive Language

Python can perform complex tasks using a few lines of code. A simple example, the hello world program you simply type `print("Hello World")`. It will take only one line to execute, while Java or C takes multiple lines.

3) Interpreted Language

Python is an interpreted language; it means the Python program is executed one line at a time. The advantage of being interpreted language, it makes debugging easy and portable.

4) Cross-platform Language

Python can run equally on different platforms such as Windows, Linux, UNIX, and Macintosh, etc. So, we can say that Python is a portable language. It enables programmers to develop the software for several competing platforms by writing a program only once.

5) Free and Open Source

Python is freely available for everyone. It is freely available on its official website www.python.org. It has a large community across the world that is dedicatedly working towards make new python modules and functions. Anyone can contribute to the Python community. The open-source means, "Anyone can download its source code without paying any penny."

6) Object-Oriented Language

Python supports object-oriented language and concepts of classes and objects come into existence. It supports inheritance, polymorphism, and encapsulation, etc. The object-oriented procedure helps to programmer to write reusable code and develop applications in less code.

7) Extensible

It implies that other languages such as C/C++ can be used to compile the code and thus it can be used further in our Python code. It converts the program into byte code, and any platform can use that byte code.

8) Large Standard Library

It provides a vast range of libraries for the various fields such as machine learning, web developer, and also for the scripting. There are various machine learning libraries, such as Tensor flow, Pandas, Numpy, Keras, and Pytorch, etc. Django, flask, pyramids are the popular framework for Python web development.

9) GUI Programming Support

Graphical User Interface is used for the developing Desktop application. PyQT5, Tkinter, Kivy are the libraries which are used for developing the web application.

10) Integrated

It can be easily integrated with languages like C, C++, and JAVA, etc. Python runs code line by line like C, C++ Java. It makes easy to debug the code.

11. Embeddable

The code of the other programming language can use in the Python source code. We can use Python source code in another programming language as well. It can embed other language into our code.

12. Dynamic Memory Allocation

In Python, we don't need to specify the data-type of the variable. When we assign some value to the variable, it automatically allocates the memory to the variable at run time. Suppose we are assigned integer value 15 to x, then we don't need to write `int x = 15`. Just write `x = 15`.

Python Applications:

Python is known for its general-purpose nature that makes it applicable in almost every domain of software development. Python makes its presence in every emerging field. It is the fastest-growing programming language and can develop any application.

Python is a general-purpose, popular programming language, and it is used in almost every technical field. The various areas of Python use are given below.

- **Data Science:** Data Science is a vast field, and Python is an important language for this field because of its simplicity, ease of use, and availability of powerful data analysis and visualization libraries like NumPy, Pandas, and Matplotlib.
- **Desktop Applications:** PyQt and Tkinter are useful libraries that can be used in GUI - Graphical User Interface-based Desktop Applications. There are better languages for this field, but it can be used with other languages for making Applications.
- **Console-based Applications:** Python is also commonly used to create command-line or console-based applications because of its ease of use and support for advanced features such as input/output redirection and piping.
- **Mobile Applications:** While Python is not commonly used for creating mobile

applications, it can still be combined with frameworks like Kivy or BeeWare to create cross-platform mobile applications.

- **Software Development:** Python is considered one of the best software-making languages. Python is easily compatible with both from Small Scale to Large Scale software.
- **Artificial Intelligence:** AI is an emerging Technology, and Python is a perfect language for artificial intelligence and machine learning because of the availability of powerful libraries such as TensorFlow, Keras, and PyTorch.
- **Web Applications:** Python is commonly used in web development on the backend with frameworks like Django and Flask and on the front end with tools like JavaScript and HTML.
- **Enterprise Applications:** Python can be used to develop large-scale enterprise applications with features such as distributed computing, networking, and parallel processing.
- **3D CAD Applications:** Python can be used for 3D computer-aided design (CAD) applications through libraries such as Blender.
- **Machine Learning:** Python is widely used for machine learning due to its simplicity, ease of use, and availability of powerful machine learning libraries.
- **Computer Vision or Image Processing Applications:** Python can be used for computer vision and image processing applications through powerful libraries such as OpenCV and Scikit-image.
- **Speech Recognition:** Python can be used for speech recognition applications through libraries such as SpeechRecognition and PyAudio.
- **Scientific computing:** Libraries like NumPy, SciPy, and Pandas provide advanced numerical computing capabilities for tasks like data analysis, machine learning, and more.
- **Education:** Python's easy-to-learn syntax and availability of many resources make it an ideal language for teaching programming to beginners.
- **Testing:** Python is used for writing automated tests, providing frameworks like unit tests and pytest that help write test cases and generate reports.

- Gaming: Python has libraries like Pygame, which provide a platform for developing games using Python.
- IoT: Python is used in IoT for developing scripts and applications for devices like Raspberry Pi, Arduino, and others.
- Networking: Python is used in networking for developing scripts and applications for network automation, monitoring, and management.
- DevOps: Python is widely used in DevOps for automation and scripting of infrastructure management, configuration management, and deployment processes.
- Finance: Python has libraries like Pandas, Scikit-learn, and Statsmodels for financial modeling and analysis.
- Audio and Music: Python has libraries like Pyaudio, which is used for audio processing, synthesis, and analysis, and Music21, which is used for music analysis and generation.
- Writing scripts: Python is used for writing utility scripts to automate tasks like file operations, web scraping, and data processing.

Here, we are specifying application areas where Python can be applied.



Fig.5.3.3 Python Applications

1) Web Applications

We can use Python to develop web applications. It provides libraries to handle internet protocols such as HTML and XML, JSON, Email processing, request, BeautifulSoup, Feedparser, etc. One of Python web-framework named Django is used on Instagram. Python provides many useful frameworks, and these are given below:

- Django and Pyramid framework(Use for heavy applications)
- Flask and Bottle (Micro-framework)
- Plone and Django CMS (Advance Content management)

2) Desktop GUI Applications

The GUI stands for the Graphical User Interface, which provides a smooth interaction to any application. Python provides a Tk GUI library to develop a user interface. Some popular GUI libraries are given below

- Tkinter or Tk
- wxWidgetM
- Kivy (used for writing multitouch applications)
- PyQt or Pyside

3) Console-based Application

Console-based applications run from the command-line or shell. These applications are computer program which are used commands to execute. This kind of application was more popular in the old generation of computers. Python can develop this kind of application very effectively. It is famous for having REPL, which means the Read-Eval-Print Loop that makes it the most suitable language for the command-line applications.

Python provides many free library or module which helps to build the command-line apps. The necessary IO libraries are used to read and write. It helps to parse argument and create console help text out-of-the-box. There are also advance libraries that can develop independent console apps.

4) Software Development

Python is useful for the software development process. It works as a support language and can be used to build control and management, testing, etc.

- SCons is used to build control.
- Buildbot and Apache Gumps are used for automated continuous compilation and testing.
- Round or Trac for bug tracking and project management.

5) Scientific and Numeric

This is the era of Artificial intelligence where the machine can perform the task the same as the human. Python language is the most suitable language for Artificial intelligence or machine learning. It consists of many scientific and mathematical libraries, which makes easy to solve complex calculations.

Implementing machine learning algorithms require complex mathematical calculation. Python has many libraries for scientific and numeric such as Numpy, Pandas, Scipy, Scikit-learn, etc. If you have some basic knowledge of Python, you need to import libraries on the top of the code. Few popular frameworks of machine libraries are given below.

- SciPy
- Scikit-learn
- NumPy
- Pandas
- Matplotlib

6) Business Applications

Business Applications differ from standard applications. E-commerce and ERP are an example of a business application. This kind of application requires extensively, scalability and readability, and Python provides all these features.

Oddo is an example of the all-in-one Python-based application which offers a range of business applications. Python provides a Tryton platform which is used to develop the business application.

7) Audio or Video-based Applications

Python is flexible to perform multiple tasks and can be used to create multimedia applications. Some multimedia applications which are made by using Python are TimPlayer, cplay, etc. These are powerful yet easy-to-use Python library for developing games and other visually rich applications on Windows, macOS, and Linux. The few multimedia libraries are given below.

- Gstreamer
- Pyglet
- QT Phonon
- Imutils

8) 3D CAD Applications

The CAD (Computer-aided design) is used to design engineering related architecture. It is used to develop the 3D representation of a part of a system. Python can create a 3D CAD application by using the following functionalities.

- Fandango (Popular)
- CAMVOX
- HeeksCNC
- AnyCAD
- RCAM

9) Enterprise Applications

Python can be used to create applications that can be used within an Enterprise or an Organization. Some real-time applications are OpenERP, Tryton, Picalo, etc.

10) Image Processing Application

Python contains many libraries that are used to work with the image. The image can be manipulated according to our requirements. Some libraries of image processing are given below.

- OpenCV
- Pillow
- SimpleITK

In this topic, we have described all types of applications where Python plays an essential role in the development of these applications. In the next tutorial, we will learn more concepts about Python.

5.3.4 WHAT IS MACHINE LEARNING?

Machine learning is a branch of artificial intelligence (AI) and computer science which focuses on the use of data and algorithms to imitate the way that humans learn, gradually improving its accuracy.

IBM has a rich history with machine learning. One of its own, Arthur Samuel, is credited for coining the term, “machine learning” with his research (link resides outside ibm.com) around the game of checkers. Robert Nealey, the self-proclaimed checkers master, played the game on an IBM 7094 computer in 1962, and he lost to the computer. Compared to what can be done today, this feat seems trivial, but it’s considered a major milestone in the field of artificial intelligence.

Over the last couple of decades, the technological advances in storage and processing power have enabled some innovative products based on machine learning, such as Netflix’s recommendation engine and self-driving cars.

Machine learning is an important component of the growing field of data science. Through the use of statistical methods, algorithms are trained to make classifications or predictions, and to uncover key insights in data mining projects. These insights subsequently drive decision making within applications and businesses, ideally impacting key growth metrics. As big data continues to expand and grow, the market demand for data scientists will increase. They will be required to help identify the most relevant business questions and the data to answer them.

Machine learning algorithms are typically created using frameworks that accelerate solution development, such as TensorFlow and PyTorch.

How machine learning works:

1. A Decision Process: In general, machine learning algorithms are used to make a prediction or classification. Based on some input data, which can be labeled or unlabelled, your algorithm will produce an estimate about a pattern in the data.
2. An Error Function: An error function evaluates the prediction of the model. If there are known examples, an error function can make a comparison to assess the accuracy of the model.
3. A Model Optimization Process: If the model can fit better to the data points in the training set, then weights are adjusted to reduce the discrepancy between the known example and the model estimate. The algorithm will repeat this “evaluate and optimize” process, updating weights autonomously until a threshold of accuracy has been met.

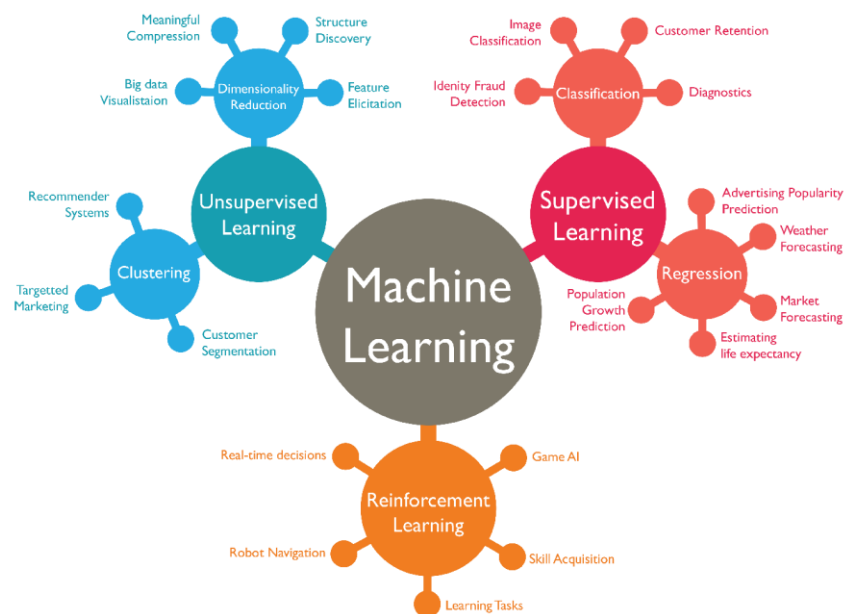


Fig.5.3.4 Introduction to Machine Learning

Machine learning methods

Machine learning models fall into three primary categories.

Supervised machine learning

Supervised learning, also known as supervised machine learning, is defined by its use of labelled datasets to train algorithms to classify data or predict outcomes accurately. As input data is fed into the model, the model adjusts its weights until it has been fitted appropriately. This occurs as part of the cross validation process to ensure that the model avoids overfitting or underfitting. Supervised learning helps organizations solve a variety of real-world problems at scale, such as classifying spam in a separate folder from your inbox. Some methods used in supervised learning include neural networks, linear regression, logistic regression, random forest, and support vector machine (SVM).

Unsupervised machine learning

Unsupervised learning, also known as unsupervised machine learning, uses machine learning algorithms to analyse and cluster unlabelled datasets. These algorithms discover hidden patterns or data groupings without the need for human intervention. This method's ability to discover similarities and differences in information make it ideal for exploratory data analysis, cross-selling strategies, customer segmentation, and image and pattern recognition. It's also used to reduce the number of features in a model through the process of dimensionality reduction. Principal component analysis (PCA) and singular value decomposition (SVD) are two common approaches for this. Other algorithms used in unsupervised learning include neural networks, k-means clustering, and probabilistic clustering methods.

Semi-supervised learning

Semi-supervised learning offers a happy medium between supervised and unsupervised learning. During training, it uses a smaller labelled data set to guide classification and feature extraction from a larger, unlabelled data set. Semi-supervised learning can solve the problem of not having enough labelled data for a supervised learning algorithm. It also helps if it's too costly to label enough data.

Common machine learning algorithms

A number of machine learning algorithms are commonly used. These include:

- **Neural networks:** Neural networks simulate the way the human brain works, with a huge number of linked processing nodes. Neural networks are good at recognizing patterns and play an important role in applications including natural language translation, image recognition, speech recognition, and image creation.
- **Linear regression:** This algorithm is used to predict numerical values, based on a linear relationship between different values. For example, the technique could be used to predict house prices based on historical data for the area.
- **Logistic regression:** This supervised learning algorithm makes predictions for categorical response variables, such as “yes/no” answers to questions. It can be used for applications such as classifying spam and quality control on a production line.
- **Clustering:** Using unsupervised learning, clustering algorithms can identify patterns in data so that it can be grouped. Computers can help data scientists by identifying differences between data items that humans have overlooked.
- **Decision trees:** Decision trees can be used for both predicting numerical values (regression) and classifying data into categories. Decision trees use a branching sequence of linked decisions that can be represented with a tree diagram. One of the advantages of decision trees is that they are easy to validate and audit, unlike the black box of the neural network.
- **Random forests:** In a random forest, the machine learning algorithm predicts a value or category by combining the results from a number of decision trees.

5.3.5 WHAT IS DEEP LEARNING?

Deep learning is a method in artificial intelligence (AI) that teaches computers to process data in a way that is inspired by the human brain. Deep learning models can recognize complex patterns in pictures, text, sounds, and other data to produce accurate insights and predictions. You can use deep learning methods to automate tasks that typically require human intelligence, such as

describing images or transcribing a sound file into text.

Why is deep learning important?

Artificial intelligence (AI) attempts to train computers to think and learn as humans do. Deep learning technology drives many AI applications used in everyday products, such as the following:

- Digital assistants
- Voice-activated television remotes
- Fraud detection
- Automatic facial recognition

It is also a critical component of emerging technologies such as self-driving cars, virtual reality, and more.

Deep learning models are computer files that data scientists have trained to perform tasks using an algorithm or a predefined set of steps. Businesses use deep learning models to analyse data and make predictions in various applications.

ACTIVATION FUNCTION	PLOT	EQUATION	DERIVATIVE	RANGE
Linear		$f(x) = x$	$f'(x) = 1$	$(-\infty, \infty)$
Binary Step		$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$	$f'(x) = \begin{cases} 0 & \text{if } x \neq 0 \\ \text{undefined} & \text{if } x = 0 \end{cases}$	$\{0, 1\}$
Sigmoid		$f(x) = \sigma(x) = \frac{1}{1 + e^{-x}}$	$f'(x) = f(x)(1 - f(x))$	$(0, 1)$
Hyperbolic Tangent(tanh)		$f(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$	$f'(x) = 1 - f(x)^2$	$(-1, 1)$
Rectified Linear Unit(ReLU)		$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$	$f'(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x > 0 \\ \text{undefined} & \text{if } x = 0 \end{cases}$	$[0, \infty)$
Softplus		$f(x) = \ln(1 + e^x)$	$f'(x) = \frac{1}{1 + e^{-x}}$	$(0, 1)$
Leaky ReLU		$f(x) = \begin{cases} 0.01x & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$	$f'(x) = \begin{cases} 0.01 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0 \end{cases}$	$(-1, 1)$
Exponential Linear Unit(ELU)		$f(x) = \begin{cases} \alpha(e^x - 1) & \text{if } x \leq 0 \\ x & \text{if } x > 0 \end{cases}$	$f'(x) = \begin{cases} \alpha e^x & \text{if } x < 0 \\ 1 & \text{if } x > 0 \\ 1 & \text{if } x = 0 \text{ and } \alpha = 1 \end{cases}$	$[0, \infty)$

Fig.5.3.5 Deep Learning Models

What are the uses of deep learning?

Deep learning has several use cases in automotive, aerospace, manufacturing, electronics, medical research, and other fields. These are some examples of deep learning:

- Self-driving cars use deep learning models to automatically detect road signs and pedestrians.
- Defence systems use deep learning to automatically flag areas of interest in satellite images.
- Medical image analysis uses deep learning to automatically detect cancer cells for medical diagnosis.
- Factories use deep learning applications to automatically detect when people or objects are within an unsafe distance of machines.

You can group these various use cases of deep learning into four broad categories—computer vision, speech recognition, natural language processing (NLP), and recommendation engines.

Computer vision

Computer vision is the computer's ability to extract information and insights from images and videos. Computers can use deep learning techniques to comprehend images in the same way that humans do. Computer vision has several applications, such as the following:

- Content moderation to automatically remove unsafe or inappropriate content from image and video archives
- Facial recognition to identify faces and recognize attributes like open eyes, glasses, and facial hair
- Image classification to identify brand logos, clothing, safety gear, and other image details

Speech recognition

Deep learning models can analyze human speech despite varying speech patterns, pitch, tone, language, and accent. Virtual assistants such as Amazon Alexa and automatic transcription software use speech recognition to do the following tasks:

- Assist call center agents and automatically classify calls.
- Convert clinical conversations into documentation in real time.
- Accurately subtitle videos and meeting recordings for a wider content reach.

Natural language processing

Computers use deep learning algorithms to gather insights and meaning from text data and documents. This ability to process natural, human-created text has several use cases, including in these functions:

- Automated virtual agents and chatbots
- Automatic summarization of documents or news articles
- Business intelligence analysis of long-form documents, such as emails and forms
- Indexing of key phrases that indicate sentiment, such as positive and negative comments on social media

Recommendation engines

Applications can use deep learning methods to track user activity and develop personalized recommendations. They can analyse the behaviour of various users and help them discover new products or services. For example, many media and entertainment companies, such as Netflix, Fox, and Peacock, use deep learning to give personalized video recommendations.

How does deep learning work?

Deep learning algorithms are neural networks that are modelled after the human brain. For example, a human brain contains millions of interconnected neurons that work together to learn and process information. Similarly, deep learning neural networks, or artificial neural networks, are made of many layers of artificial neurons that work together inside the computer.

Artificial neurons are software modules called nodes, which use mathematical calculations to process data. Artificial neural networks are deep learning algorithms that use these nodes to solve complex problems.

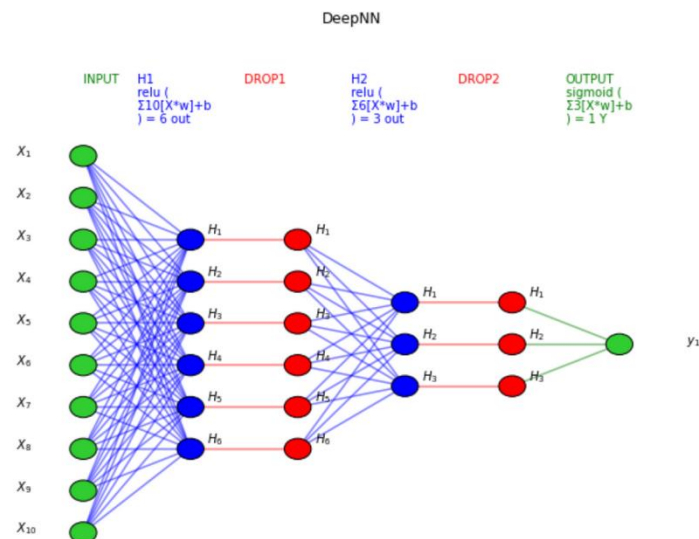


Fig. 5.3.5.1 Deep Learning Works

What are the components of a deep learning network?

The components of a deep neural network are the following.

Input layer

An artificial neural network has several nodes that input data into it. These nodes make up the input layer of the system.

Hidden layer

The input layer processes and passes the data to layers further in the neural network. These hidden layers process information at different levels, adapting their behavior as they receive new information. Deep learning networks have hundreds of hidden layers that they can use to analyze a problem from several different angles.

For example, if you were given an image of an unknown animal that you had to classify, you would compare it with animals you already know. For example, you would look at the shape of its eyes and ears, its size, the number of legs, and its fur pattern. You would try to identify patterns, such as the following:

- The animal has hooves, so it could be a cow or deer.
- The animal has cat eyes, so it could be some type of wild cat.

The hidden layers in deep neural networks work in the same way. If a deep learning algorithm is trying to classify an animal image, each of its hidden layers processes a different feature of the animal and tries to accurately categorize it.

Output layer

The output layer consists of the nodes that output the data. Deep learning models that output "yes" or "no" answers have only two nodes in the output layer. On the other hand, those that output a wider range of answers have more nodes.

What is deep learning in the context of machine learning?

Deep learning is a subset of machine learning. Deep learning algorithms emerged in an attempt to make traditional machine learning techniques more efficient. Traditional machine learning methods require significant human effort to train the software. For example, in animal image recognition, you need to do the following:

- Manually label hundreds of thousands of animal images.

- Make the machine learning algorithms process those images.
- Test those algorithms on a set of unknown images.
- Identify why some results are inaccurate.
- Improve the dataset by labeling new images to improve result accuracy.

This process is called supervised learning. In supervised learning, result accuracy improves only when you have a broad and sufficiently varied dataset. For instance, the algorithm might accurately identify black cats but not white cats because the training dataset had more images of black cats. In that case, you would need to label more white cat images and train the machine learning models once again.

What are the benefits of deep learning over machine learning?

A deep learning network has the following benefits over traditional machine learning.

Efficient processing of unstructured data

Machine learning methods find unstructured data, such as text documents, challenging to process because the training dataset can have infinite variations. On the other hand, deep learning models can comprehend unstructured data and make general observations without manual feature extraction. For instance, a neural network can recognize that these two different input sentences have the same meaning:

- Can you tell me how to make the payment?
- How do I transfer money?

Hidden relationships and pattern discovery

A deep learning application can analyse large amounts of data more deeply and reveal new insights for which it might not have been trained. For example, consider a deep learning model that is trained to analyse consumer purchases. The model has data only for the items you have

already purchased. However, the artificial neural network can suggest new items that you haven't bought by comparing your buying patterns to those of other similar customers.

Unsupervised learning

Deep learning models can learn and improve over time based on user behaviour. They do not require large variations of labelled datasets. For example, consider a neural network that automatically corrects or suggests words by analysing your typing behaviour. Let's assume it was trained in the English language and can spell-check English words. However, if you frequently type non-English words, such as *danke*, the neural network automatically learns and autocorrects these words too.

Volatile data processing

Volatile datasets have large variations. One example is loan repayment amounts in a bank. A deep learning neural network can categorize and sort that data as well, such as by analysing financial transactions and flagging some of them for fraud detection.

What are the challenges of deep learning?

As deep learning is a relatively new technology, certain challenges come with its practical implementation.

Large quantities of high-quality data

Deep learning algorithms give better results when you train them on large amounts of high-quality data. Outliers or mistakes in your input dataset can significantly affect the deep learning process. For instance, in our animal image example, the deep learning model might classify an airplane as a turtle if non-animal images were accidentally introduced in the dataset.

To avoid such inaccuracies, you must clean and process large amounts of data before you can train deep learning models. The input data pre-processing requires large amounts of data storage capacity.

Large processing power

Deep learning algorithms are compute-intensive and require infrastructure with sufficient compute capacity to properly function. Otherwise, they take a long time to process results.

CHAPTER 6

MODULES DESCRIPTION

- Data Acquisition Module
- Data Preprocessing Module
- Feature Extraction Module
- CNN Model Training Module
- AF Detection / Classification Module
- Result Display and Monitoring Module

6.1 DATA ACQUISITION MODULE

- Collects ECG signals from wearable devices or ECG datasets.
- Provides real-time heart activity data for analysis.
- Sampling frequency: 200 Hz or higher

6.2 DATA PREPROCESSING MODULE

- Noise removal using digital filters
- Baseline wander correction
- Signal segmentation
- Normalization

6.3 FEATURE EXTRACTION MODULE

ECG signals are converted into 2D spectrograms using Short-Time Fourier Transform (STFT) to capture time-frequency information.

- Converts ECG signals into **2D spectrogram or wavelet images**.
- Extracts important time and frequency features from ECG data.

6.4 CNN MODEL TRAINING MODULE

1.The CNN architecture includes:

- Convolutional layers for feature extraction

2.ReLU activation functions

- Max-pooling layers
- Fully connected layers
- Softmax output layer

3. The model learns:

- Absence of P-wave
- Irregular RR intervals
- Atrial signal abnormalities
 - Trains the **CNN model** using labeled ECG datasets.
 - Helps the system learn patterns of normal and AF signals.

6.5 AF DETECTION / CLASSIFICATION MODULE

- Classifies ECG signals as Normal or Atrial Fibrillation.
- Uses the trained CNN model for accurate detection.

6.6 RESULT DISPLAY AND MONITORING MODULE

- Dataset splitting: 70% training, 15% validation, 15% testing
- Data augmentation
- Cross-validation
- Adam optimizer
- Binary cross-entropy loss

CHAPTER 7

EXPERIMENTAL RESULTS

The model was evaluated using:

- Accuracy
- Sensitivity
- Specificity
- Precision
- F1-score

Results demonstrate:

- High classification accuracy
- Improved sensitivity compared to ANN
- Reduced false positives
- Better generalization
- CNN-based model outperformed traditional SVM and MLP models.

DISCUSSIONS

The proposed CNN-based approach:

- Eliminates manual feature extraction
- Improves detection reliability
- Enables real-time deployment
- Supports wearable and mHealth integration

CHALLENGES:

- Large dataset requirement
- Computational cost
- Need for cross-population validation

CHAPTER 8

SYSTEM TESTING

Testing of Product

System testing is the stage of implementation, which aimed at ensuring that system works accurately and efficiently before the live operation commence. Testing is the process of executing a program with the intent of finding an error. A good test case is one that has a high probability of finding an error. A successful test is one that answers a yet undiscovered error.

Testing is vital to the success of the system. System testing makes a logical assumption that if all parts of the system are correct, the goal will be successfully achieved. The candidate system is subject to variety of tests-on-line response, Volume Street, recovery and security and usability test. A series of tests are performed before the system is ready for the user acceptance testing. Any engineered product can be tested in one of the following ways. Knowing the specified function that a product has been designed to from, test can be conducted to demonstrate each function is fully operational. Knowing the internal working of a product, tests can be conducted to ensure that “al gears mesh”, that is the internal operation of the product performs according to the specification and all internal components have been adequately exercised.

UNIT TESTING

Unit testing is the testing of each module and the integration of the overall system is done. Unit testing becomes verification efforts on the smallest unit of software design in the module. This is also known as ‘module testing’. The

modules of the system are tested separately. This testing is carried out during the programming itself. In this testing step, each model is found to be working satisfactorily as regard to the expected output from the module. There are some validation checks for the fields. For example, the validation check is done for verifying the data given by the user where both format and validity of the data entered is included. It is very easy to find error and debug the system.

INTEGRATION TESTING

Data can be lost across an interface, one module can have an adverse effect on the other sub function, when combined, may not produce the desired major function. Integrated testing is systematic testing that can be done with sample data. The need for the integrated test is to find the overall system performance. There are two types of integration testing. They are:

- i) Top-down integration testing.
- ii) Bottom-up integration testing.

WHITE BOX TESTING

White Box testing is a test case design method that uses the control structure of the procedural design to drive cases. Using the white box testing methods, we derived test cases that guarantee that all independent paths within a module have been exercised at least once.

BLACK BOX TESTING

- ✓ Black box testing is done to find incorrect or missing function
- ✓ Interface error
- ✓ Errors in external database access

- ✓ Performance errors
- ✓ Initialization and termination errors

In 'functional testing', is performed to validate an application conforms to its specifications of correctly performs all its required functions. So this testing is also called 'black box testing'. It tests the external behaviour of the system. Here the engineered product can be tested knowing the specified function that a product has been designed to perform, tests can be conducted to demonstrate that each function is fully operational.

VALIDATION TESTING

After the culmination of black box testing, software is completed assembly as a package, interfacing errors have been uncovered and corrected and final series of software validation tests begin validation testing can be defined as many, but a single definition is that validation succeeds when the software functions in a manner that can be reasonably expected by the customer.

USER ACCEPTANCE TESTING

User acceptance of the system is the key factor for the success of the system. The system under consideration is tested for user acceptance by constantly keeping in touch with prospective system at the time of developing changes whenever required.

Penetration Testing

It is a type of security testing, also known as pen test in short. Penetration testing is done to test how secure software and its environments (Hardware, Operating system and network) are when subject to attack by an external or internal intruder. Intruder can be a human/hacker or malicious programs. Pen test uses methods to forcibly intrude (by brute force attack) or by using a weakness (vulnerability) to gain access to a software or data or hardware with an intent to expose ways to steal, manipulate or corrupt data, software files or configuration. Penetration Testing is a way of ethical hacking, an experienced Penetration tester will use the same methods and tools that a hacker would use but the intention of Penetration tester is to identify vulnerability and get them fixed before a real hacker or malicious program exploits it.

Regression Testing

It is a type of software testing that is carried out by software testers as functional regression tests and developers as Unit regression tests. Objective of regression tests are to find defects that got introduced to defect fix (is) or introduction of new feature(s). Regression tests are ideal candidate for automation.

Retesting

It is a type of retesting that is carried out by software testers as a part of defect fix verification. For e.g. a tester is verifying a defect fix and let us say that there are 3 test cases failed due to this defect. Once tester verifies defect fix as resolved, test will retest or test the same functionality again by executing the test cases that were failed earlier.

Risk based Testing

It is a type of software testing and a different approach towards testing a software. In Risk based testing, requirements and functionality of software to be tested are prioritized as Critical, High, Medium and low. In this approach, all critical and high priority tests are tested and then followed by Medium. Low priority or low risk functionality are tested at the end or may not base on the time available for testing.

Smoke testing

It is a type of testing that is carried out by software testers to check if the new build provided by development team is stable enough i.e., major functionality is working as expected in order to carry out further or detailed testing. Smoke testing is intended to find “show stopper” defects that can prevent testers from testing the application in detail. Smoke testing carried out for a build is also known as build verification test.

Security Testing

It is a type of software testing carried out by specialized team of software testers. Objective of security testing is to secure the software is to external or internal threats from humans and malicious programs. Security testing basically checks, how good is software’s authorization mechanism, how strong is authentication, how software maintains confidentiality of the data, how does the software maintain integrity of the data, what is the availability of the software in an event of an attack on the software by hackers and malicious programs is for Security testing requires good knowledge of application, technology, networking,

security testing tools. With increasing number of web applications necessarily of security testing has increased to a greater extent.

Sanity Testing

It is a type of testing that is carried out mostly by testers and in some projects by developers as well. Sanity testing is a quick evaluation of the software, environment, network, external systems are up & running, software environment as a whole is stable enough to proceed with extensive testing. Sanity tests are narrow and most of the time sanity tests are not documented.

Scalability Testing

It is a non-functional test intended to test one of the software quality attributes i.e. “Scalability”. Scalability test is not focused on just one or few functionality of the software instead performance of software as a whole. Scalability testing is usually done by performance engineering team. Objective of scalability testing is to test the ability of the software to scale up with increased users, increased transactions, increase in database size etc., It is not necessary that software’s performance increases with increase in hardware configuration, scalability tests helps to find out how much more workload the software can support with expanding user base, transactions, data storage etc.,

Stability Testing

It is a non-functional test intended to test one of the software quality attributes i.e. “Stability”. Stability testing focuses on testing how stable software is when it is subject to loads at acceptable levels, peak loads, loads generated in spikes, with more volumes of data to be processed. Scalability testing will involve

performing different types of performance tests like load testing, stress testing, spike testing, soak testing, spike testing etc...

Static Testing is a form of testing where in approaches like reviews, walkthroughs are employed to evaluate the correctness of the deliverable. In static testing software code is not executed instead it is reviewed for syntax, commenting, naming convention, size of the functions and methods etc. Static testing usually has check lists against which deliverables are evaluated. Static testing can be applied for requirements, designs, and test cases by using approaches like reviews or walkthroughs.

Stress Testing is a type of performance testing, in which software is subjected to peak loads and even to a break point to observe how the software would behave at breakpoint. Stress testing also tests the behaviour of the software with insufficient resources like CPU, Memory, Network bandwidth, Disk space etc. Stress testing enables to check some of the quality attributes like robustness and reliability.

CHAPTER 9

CONCLUSION

In conclusion, the proposed system for Atrial Fibrillation detection using Convolutional Neural Networks represents a significant advancement in the field of cardiac monitoring and diagnosis. By harnessing the capabilities of deep learning and real-time data processing, the system offers a robust solution for identifying AF episodes with high accuracy. The integration of wearable technology allows for continuous monitoring of patients, enabling timely interventions that can significantly reduce the risk of severe complications associated with AF.

The literature reviewed highlights the growing potential of machine learning approaches in improving AF detection, but also underscores the challenges of data quality and model generalization across diverse populations. The proposed system addresses these challenges through rigorous training and validation processes, ensuring that the model is not only accurate but also applicable in varied clinical environments. As the healthcare landscape evolves towards more personalized and technology-driven approaches, this system stands to play a critical role in enhancing patient care and outcomes in the management of Atrial Fibrillation. Future work will focus on expanding the dataset for training, refining the model further, and exploring integration with broader health monitoring systems to maximize its impact in clinical practice.

9.1 FUTURE WORK

- Integrating the system with wearable devices and smartwatches for continuous heart monitoring.
- Improving the model accuracy using larger ECG datasets and advanced deep learning algorithms.
- Developing a mobile application to display ECG results and AF alerts in real time.
- Implementing cloud-based storage for secure data management and remote monitoring by doctors.
- Expanding the system to detect other heart diseases and arrhythmias in addition to AF.

APPENDIX

A1. SAMPLE CODING

[views.py](#)

```
from django.shortcuts import render
from myapp.models import Uploads
from keras.models import load_model
import numpy as np
import os
import matplotlib

import matplotlib.pyplot as plt
matplotlib.use('Agg')

# Create your views here.
def index(request):
    if request.method == 'POST':
        file = request.FILES['file']
        upload = Uploads(file=file)
        upload.save()
        print('File uploaded successfully')
        print('File name:', file.name)

    # Load the saved model
    model = load_model("static/cnn_model_from_train_dataset.h5")

    # Get the base directory
    base_dir = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
    print("Base directory:", upload)

    sample_path = base_dir + upload.file.url
```

```

sample = np.load(sample_path)

# Reshape the sample to match the input shape of the model
sample = sample.reshape(1, sample.shape[0], 1)

# Predict using the model
prediction = model.predict(sample)
prediction = "Abnormal" if prediction[0][0] > 0.5 else "Normal"
print("Prediction:", prediction)

# Plot the sample
# plt.figure(figsize=(10, 4))
# plt.plot(sample[0, :, 0], label="ECG Signal")
# plt.title("ECG Signal Sample")
# plt.xlabel("Time Index")
# plt.ylabel("Amplitude")
# plt.legend()
# plt.show()

# Save the plot without showing it
static_dir = os.path.join(base_dir, 'static')
os.makedirs(static_dir, exist_ok=True)

plt.figure(figsize=(10, 4))
plt.plot(sample[0, :, 0], label="ECG Signal")
plt.title("ECG Signal Sample")
plt.xlabel("Time Index")
plt.ylabel("Amplitude")
plt.legend()

# Save the plot
plot_path = os.path.join(static_dir, 'ecg_plot.png')
plt.savefig(plot_path)
plt.close() # Close figure to free memory

```

```
data={'prediction': prediction, 'plot_path': '/static/ecg_plot.png'}
return render(request, 'index.html', data)
```

```
return render(request, 'index.html')
```

Index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Upload .npy File</title>
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
rel="stylesheet">
</head>
<body>
  <div class="container mt-5">
    <h2 class="text-center">Atrial Fibrillation Detection</h2>
    <form enctype="multipart/form-data" method="post" action="/" >

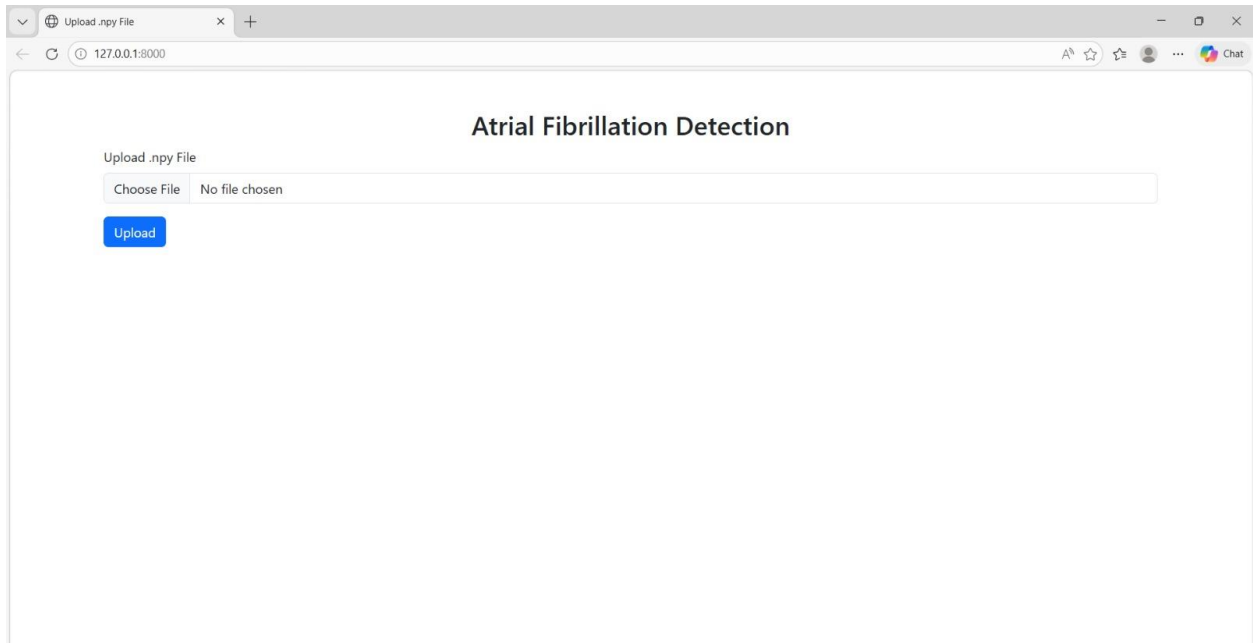
      {% csrf_token %}
      <div class="mb-3">
        <label for="fileInput" class="form-label">Upload .npy File</label>
        <input class="form-control" type="file" id="fileInput" name="file"
accept=".npy" required>
      </div>
      <button type="submit" class="btn btn-primary">Upload</button>
    </form>
    <div id="response" class="mt-3"></div>
    {% if prediction %}
      <div class="alert alert-info" role="alert">
        {{ prediction }}
      </div>
```

```
{% endif %}

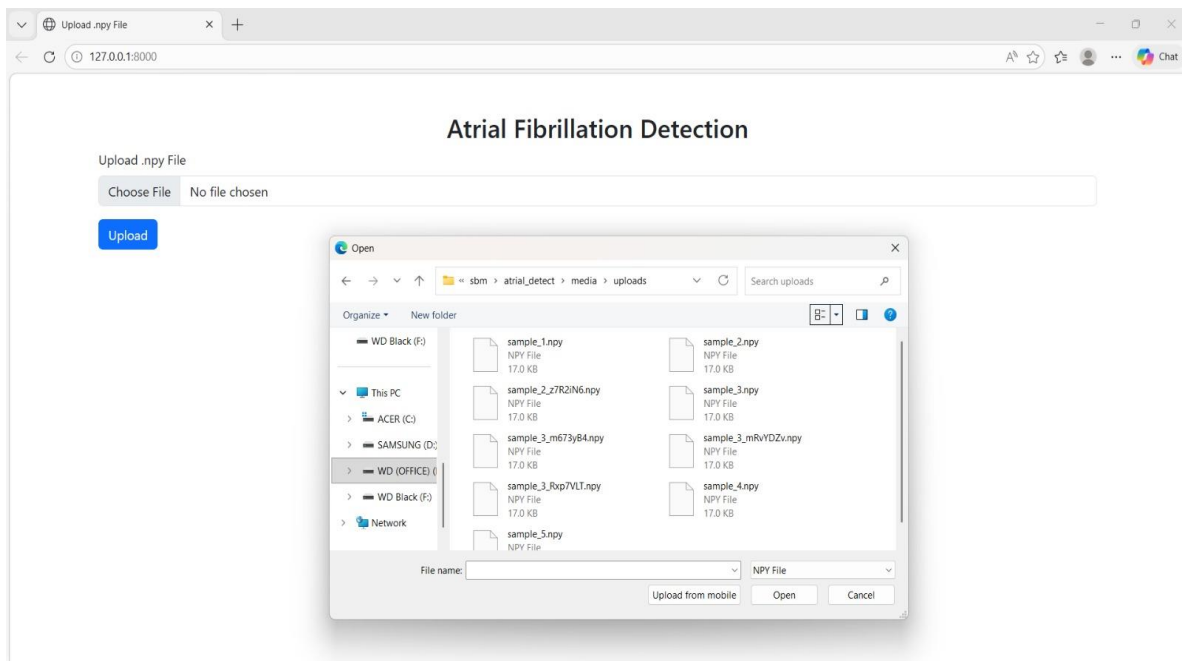
{% if plot_path %}
  <div class="text-center mt-3">
    
  </div>
{% endif %}
</div>

</body>
</html>
```

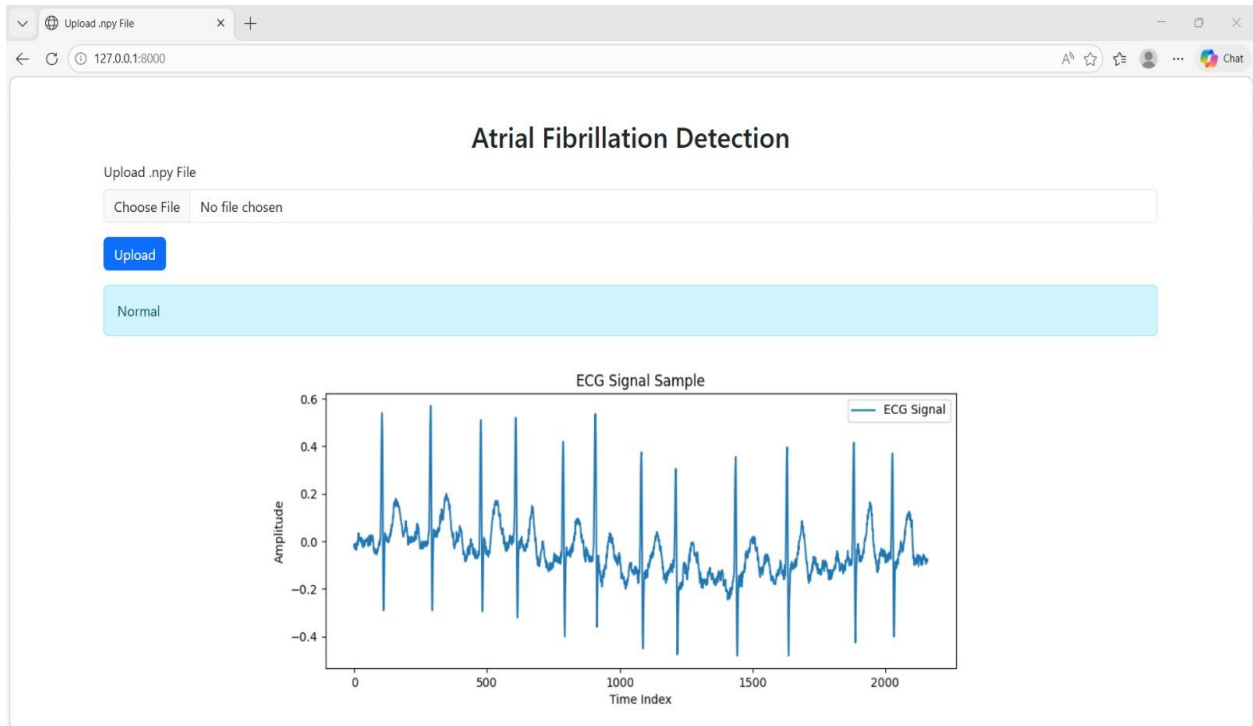
A2. SAMPLE SCREENSHOTS



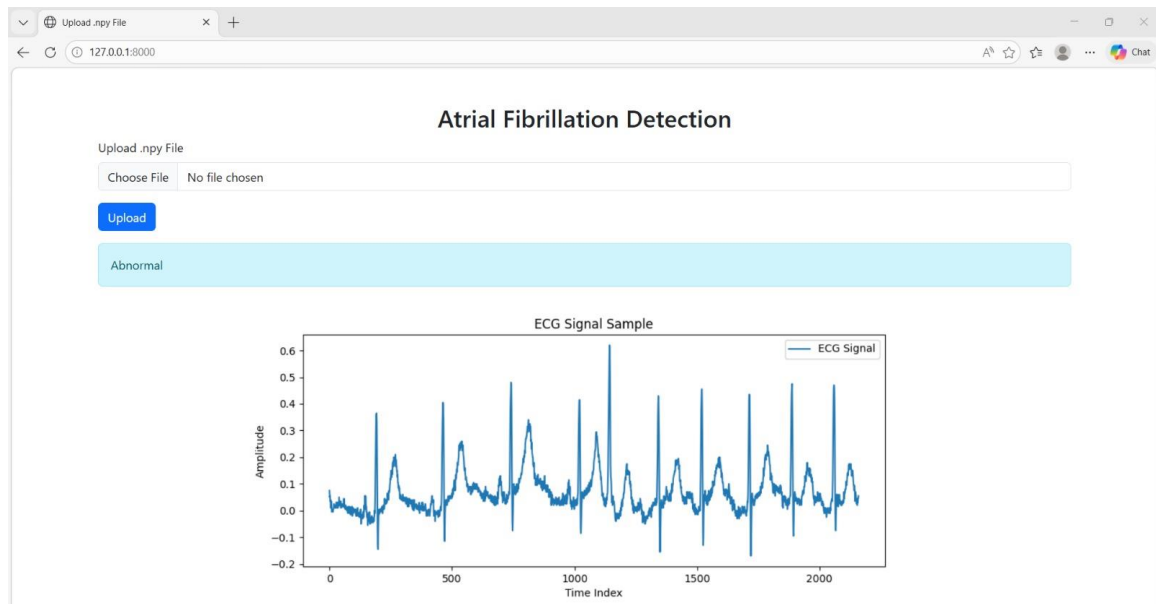
A.2.1.Atrial Fibillation Detection – Upload the file



A.2.2 Atrial Fibillation Detection – Selecting the file



A.2.3. Atrial Fibrillation Detection – Sample Result1



A.2.4 Atrial Fibrillation Detection – Sample Result2

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